Assignment 2

Name: Ridhi
Roll No: 1024030528
Batch: 3C35

Q1. Develop a Java application for an online food delivery company that manages different types of food orders using packages food.model, food.service, food.utility and food.main.

FoodOrder.java

```
package food.model;

public abstract class FoodOrder {
    private int orderId;
    private String customerName;
    private double amount;
    private static String restaurantName = "Tasty Bites";
    private static int orderCount = 0;

    public FoodOrder(int orderId, String customerName, double amount) {
        this.orderId = orderId;
        this.customerName = customerName;
        this.amount = amount;
        orderCount++;
    }

    public int getOrderId() {
        return orderId;
    }

    public void setOrderId(int orderId) {
        this.orderId = orderId;
    }

    public String getCustomerName() {
        return customerName;
    }

    public void setCustomerName(String customerName) {
        this.customerName = customerName;
    }

    public double getAmount() {
        return amount;
    }

    public void setAmount(double amount) {
        this.amount = amount;
    }

    public static String getRestaurantName() {
        return restaurantName;
    }
```

```
    public static void displayTotalOrders() {
        System.out.println("Total Orders Created: " + orderCount);
    }

    public abstract double calculateDeliveryCharge();
}
```

Discountable.java

```
package food.model;

public interface Discountable {
    double applyDiscount();
}
```

RegularOrder.java

```
package food.model;

public class RegularOrder extends FoodOrder implements Discountable {

    public RegularOrder(int orderId, String customerName, double amount) {
        super(orderId, customerName, amount);
    }

    public double calculateDeliveryCharge() {
        return 80;
    }

    public double applyDiscount() {
        return getAmount() * 0.10;
    }
}
```

PremiumOrder.java

```
package food.model;

public class PremiumOrder extends FoodOrder implements Discountable {

    public PremiumOrder(int orderId, String customerName, double amount) {
        super(orderId, customerName, amount);
    }

    public double calculateDeliveryCharge() {
        return 50;
    }

    public double applyDiscount() {
        return getAmount() * 0.15;
    }
}
```

OrderUtility.java

```
package food.utility;

import food.model.FoodOrder;
import food.model.Discountable;
```

```
public class OrderUtility {

    public static boolean validateAmount(double amount) {
        return amount > 0;
    }

    public static boolean validateCustomerName(String name) {
        return name != null && !name.trim().isEmpty();
    }

    public static void generateOrderSummary(FoodOrder order) {
        double discount = 0;
        if (order instanceof Discountable) {
            discount = ((Discountable) order).applyDiscount();
        }
        double deliveryCharge = order.calculateDeliveryCharge();
        double finalAmount = order.getAmount() - discount + deliveryCharge;

        System.out.println("Order ID: " + order.getOrderId());
        System.out.println("Customer Name: " + order.getCustomerName());
        System.out.println("Bill Amount: Rs. " + order.getAmount());
        System.out.println("Discount: Rs. " + discount);
        System.out.println("Delivery Charge: Rs. " + deliveryCharge);
        System.out.println("Final Payable Amount: Rs. " + finalAmount);
        System.out.println("----------------------------------------");
    }
}
```

Main.java

```
package food.main;

import food.model.FoodOrder;
import food.model.RegularOrder;
import food.model.PremiumOrder;
import food.utility.OrderUtility;

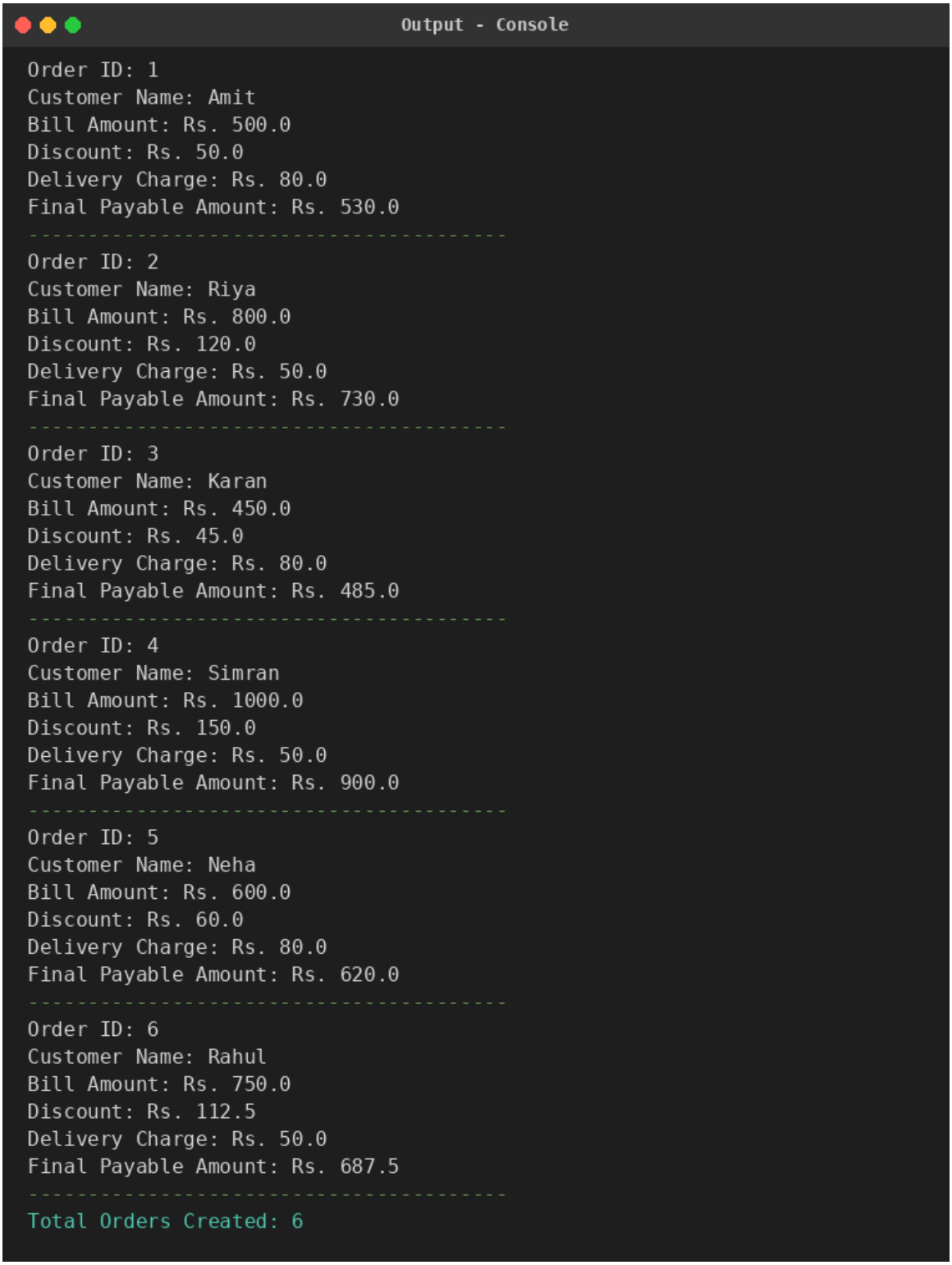
public class Main {
    public static void main(String[] args) {
        FoodOrder[] orders = new FoodOrder[6];

        orders[0] = new RegularOrder(1, "Amit", 500);
        orders[1] = new PremiumOrder(2, "Riya", 800);
        orders[2] = new RegularOrder(3, "Karan", 450);
        orders[3] = new PremiumOrder(4, "Simran", 1000);
        orders[4] = new RegularOrder(5, "Neha", 600);
        orders[5] = new PremiumOrder(6, "Rahul", 750);

        for (FoodOrder order : orders) {
            if (OrderUtility.validateAmount(order.getAmount()) &&
                OrderUtility.validateCustomerName(order.getCustomerName())) {
                OrderUtility.generateOrderSummary(order);
            }
        }
```

```
        FoodOrder.displayTotalOrders();
    }
}
```

Output



Q2. A university wants to develop a Smart Library Resource Management System to manage Books, Journals, and Digital Resources using packages library.model, library.service, library.util and library.main.

LibraryResource.java

```
package library.model;

public abstract class LibraryResource {
    private int resourceId;
    private String title;
    private String author;
    protected static String libraryName = "Central Library";
    private static int resourceCount = 0;

    public LibraryResource(int resourceId, String title, String author) {
        this.resourceId = resourceId;
        this.title = title;
        this.author = author;
        resourceCount++;
    }

    public int getResourceId() {
        return resourceId;
    }

    public String getTitle() {
        return title;
    }

    public String getAuthor() {
        return author;
    }

    protected String getBasicInfo() {
        return "Resource ID: " + resourceId + ", Title: " + title + ", Author: " + author;
    }

    public static void displayTotalResources() {
        System.out.println("Total Resources Created: " + resourceCount);
    }

    public abstract double calculateFine(int overdueDays);
}
```

Printable.java

```
package library.model;

public interface Printable {
    void printDetails();
}
```

Book.java

```
package library.model;

public class Book extends LibraryResource implements Printable {

    public Book(int resourceId, String title, String author) {
        super(resourceId, title, author);
    }
```

```
    public double calculateFine(int overdueDays) {
        return overdueDays * 5;
    }

    public void printDetails() {
        System.out.println(getBasicInfo());
        System.out.println("Type: Book");
        System.out.println("Library: " + libraryName);
    }
}
```

DigitalResource.java

```
package library.model;

public class DigitalResource extends LibraryResource implements Printable {

    public DigitalResource(int resourceId, String title, String author) {
        super(resourceId, title, author);
    }

    public double calculateFine(int overdueDays) {
        return overdueDays * 2;
    }

    public void printDetails() {
        System.out.println(getBasicInfo());
        System.out.println("Type: Digital Resource");
        System.out.println("Library: " + libraryName);
    }
}
```

InputValidator.java

```
package library.util;

public class InputValidator {

    public static boolean validateResourceId(int resourceId) {
        return resourceId > 0;
    }

    public static boolean validateFineDays(int days) {
        return days >= 0;
    }
}
```

Main.java

```
package library.main;

import library.model.LibraryResource;
import library.model.Book;
import library.model.DigitalResource;
import library.model.Printable;
import library.util.InputValidator;
```

```
public class Main {
    public static void main(String[] args) {
        LibraryResource[] resources = new LibraryResource[5];

        resources[0] = new Book(1, "Java Programming", "James Gosling");
        resources[1] = new DigitalResource(2, "Data Structures E-Book", "Robert Sedgewick");
        resources[2] = new Book(3, "Operating Systems", "Silberschatz");
        resources[3] = new DigitalResource(4, "Computer Networks PDF", "Tanenbaum");
        resources[4] = new Book(5, "Database Management", "Korth");

        int[] overdueDays = {5, 3, 0, 7, 2};
        double totalFine = 0;

        for (int i = 0; i < resources.length; i++) {
            if (!InputValidator.validateResourceId(resources[i].getResourceId())) {
                continue;
            }

            if (resources[i] instanceof Printable) {
                ((Printable) resources[i]).printDetails();
            }

            if (InputValidator.validateFineDays(overdueDays[i])) {
                double fine = resources[i].calculateFine(overdueDays[i]);
                System.out.println("Overdue Days: " + overdueDays[i]);
                System.out.println("Fine: Rs. " + fine);
                totalFine += fine;
            }
            System.out.println("----------------------------------------");
        }

        System.out.println("Total Fine of All Overdue Resources: Rs. " + totalFine);
        LibraryResource.displayTotalResources();
    }
}
```

Output


Output - Console
Resource ID: 1, Title: Java Programming, Author: James Gosling
Type: Book
Library: Central Library
Overdue Days: 5
Fine: Rs. 25.0

Resource ID: 2, Title: Data Structures E-Book, Author: Robert Sedgewick
Type: Digital Resource
Library: Central Library
Overdue Days: 3
Fine: Rs. 6.0

Resource ID: 3, Title: Operating Systems, Author: Silberschatz
Type: Book
Library: Central Library
Overdue Days: 0
Fine: Rs. 0.0

Resource ID: 4, Title: Computer Networks PDF, Author: Tanenbaum
Type: Digital Resource
Library: Central Library
Overdue Days: 7
Fine: Rs. 14.0

Resource ID: 5, Title: Database Management, Author: Korth
Type: Book
Library: Central Library
Overdue Days: 2
Fine: Rs. 10.0

Total Fine of All Overdue Resources: Rs. 55.0
Total Resources Created: 5